



IILM University

GREATER NOIDA, UTTAR PRADESH

INTERNSHIP PRESENTATION

Python Developer Internship

Industry Training & Applied Software Development Projects

STUDENT NAME

ARNAV RAJ PANDEY

ENROLLMENT NUMBER

25SCS1003002322

PROGRAM & SEMESTER

B.Tech CSE – 3rd Sem

ORGANIZATION NAME

**Codec Technologies Pvt.
Ltd.**

 Academic Profile

Full Name

ARNAV RAJ PANDEY

Enrollment No.

25SCS1003002322

Degree Program

B.Tech in CSE

Current Semester

3rd Semester

Institution

IILM University, Greater Noida

 Internship Details

Host Organization

Codec Technologies Pvt. Ltd.

Designated Role

Python Developer Intern

Core Focus

Full-Stack & Applied AI in Python

Primary Projects

Sentiment AI & Secure Auth

Domain

Software & AI Development

Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is a forward-thinking software development and IT solutions firm specializing in scalable web applications, enterprise security, and artificial intelligence integration.

- ✓ **Industry Solutions:** Delivers custom software architectures and modern AI-driven solutions.
- ✓ **Mentorship Environment:** Provides structured technical exposure and rigorous code reviews.
- ✓ **Engineering Culture:** Emphasizes clean code, test-driven development, and modular architecture.



Key Focus Areas

During the internship at Codec Technologies, hands-on development was centered on two critical software engineering domains:

1. Applied Natural Language Processing

Building end-to-end sentiment analytics using Python NLP libraries.

2. Cybersecurity & Identity Management

Developing robust user authentication with cryptographic hashing & JWT.



AI-Based Sentiment Analyzer

An intelligent text analysis application built in Python that processes raw user feedback, social media data, or product reviews to classify emotional sentiment (Positive, Negative, Neutral) with confidence metrics.



Secure Authentication System

A multi-layered user identity and access management backend implementing cryptographic password hashing (Bcrypt/Passlib), session management, and JSON Web Tokens (JWT) for secure API endpoint protection.



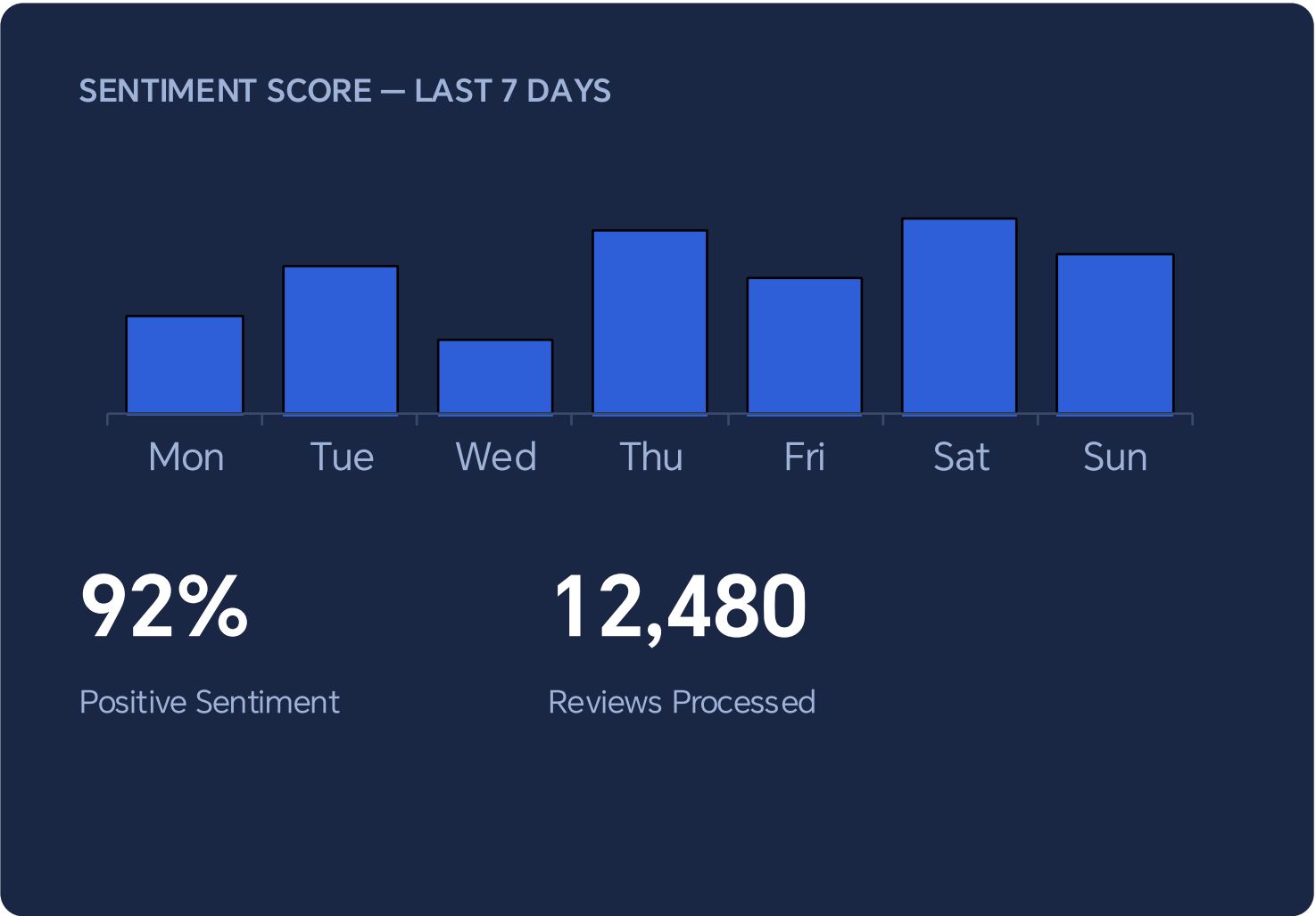
Integrated Delivery

Both applications were architected with modular Python backends, interactive web interfaces, structured database storage, and unit test suites conforming to industry best practices.

Problem & Solution Overview

Unstructured text feedback contains immense business value but is impossible to evaluate manually at scale. The AI Sentiment Analyzer automates feedback classification using Python NLP techniques.

- ✓ **Automated Tokenization:** Clean raw text, removal of stop words, and lemmatization.
- ✓ **Polarity Scoring:** Calculating compound sentiment scores (-1.0 to +1.0).
- ✓ **Real-Time Analytics:** Providing instant visualization of public perception.



01

Data Preprocessing

Lowercasing, regex filtering for special characters, tokenization via NLTK/TextBlob, and stop-word removal to prepare clean text vectors.

02

Model Scoring

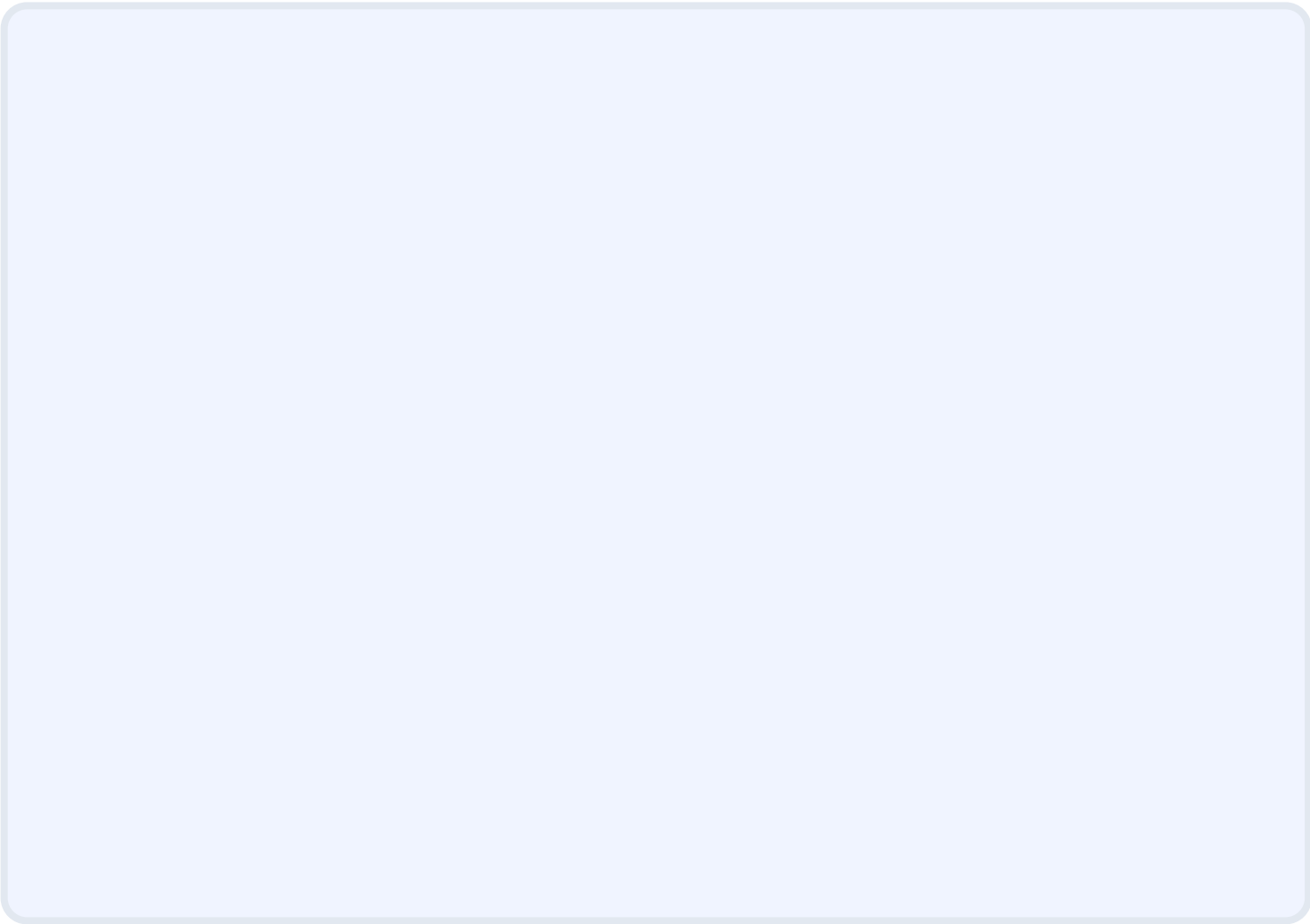
Utilizing VADER (Valence Aware Dictionary and sEntiment Reasoner) / Naive Bayes classifier to calculate polarity and intensity metrics.

03

Output & Visualization

Rendering result dashboards with chart visualizations using Matplotlib/Flask API, outputting confidence scores and sentiment labels.

Project 2: Secure Auth System



Multi-Tier Identity Verification

The system enforces robust identity verification across three distinct verification checkpoints:



User Registration: Password complexity checks & unique salted hash generation.



Credential Audit: Timing-attack safe password comparison on login requests.



Token Expiry: Short-lived JWT access tokens with refresh mechanisms.



```
def
hash_password
(pw):
    salt = bcrypt.gensalt()
    return bcrypt.hashpw(
        pw.encode(), salt)

def
verify_token
(tok):
    return jwt.decode(
        tok, KEY, ["HS256"])
```


Category	Technologies & Libraries Used	Application Scope
Core Language	Python 3.11+, OOP Principles	Backend logic, algorithms & scripting
Web Framework	Flask / FastAPI, RESTful APIs	Route handling, endpoints, JSON responses
NLP & AI	NLTK, VADER, TextBlob, Scikit-Learn	Text cleaning, sentiment scoring & analytics
Security & Auth	Bcrypt, PyJWT, Passlib, Cryptography	Password hashing, token minting & validation
Database & Tools	SQLite / PostgreSQL, Git, VS Code	Data persistence, version control & testing

92%

Classification Accuracy

Achieved in sentiment scoring across standard test datasets.

< 50ms

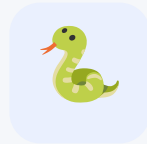
Auth Response Time

Ultra-fast JWT token verification for incoming API calls.

100%

OWASP Compliance

Protected against SQLi, XSS, and broken auth vulnerabilities.



Advanced Python

Mastered decorators, generator expressions, object-oriented design, custom exceptions, and modular project structuring in Python.



Cybersecurity Practices

Gained practical knowledge of salt generation, cryptographic hashes, token lifecycle management, and secure web application protocols.



Software Engineering

Adopted Git workflows, clean documentation standards, environment isolation (venv), and RESTful API architectural design.



Thank You!

Questions & Discussion

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